

Shanghao Liu

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Education

**Ph.D. in Computer Science**Aug 2026 (expected) - Singapore

SCHOOL OF COMPUTING, NATIONAL UNIVERSITY OF SINGAPORE

- Advisor: Prof. Wenqi Jiang

**M.S. in Computer Science**2023 - Jun 2026 (expected) Beijing

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING, BEIHANG UNIVERSITY

- Advisor: Prof. Hailong Yang | GPA: 3.52/4.00 | IELTS: 6.5
- Thesis: Performance Optimization of Sparse Attention for Video Generation
- Award: First-Class Scholarship (2023)

**B.S. in Physics**2019 - 2023 Lanzhou

SCHOOL OF PHYSICAL SCIENCE AND TECHNOLOGY, LANZHOU UNIVERSITY

- Thesis: Neural Architecture Search for Gravitational-Wave Detection in Binary Star Systems (Outstanding Thesis Award)
- GPA: 89.29/100
- Awards: First-Class Scholarship (2021), Second-Class Scholarship (2020, 2022), Huawei Kunpeng Scholarship (2021)

Publications

- (1) Liu, S.; Chen, R.; Zheng, S.; Liu, Y.; Liang, Y. (Eric); Yang, H. SPADE: An Input-Adaptive Sparse Attention Engine for Fast Video Diffusion Models Inference. In *Proceedings of the the Chips to Systems Conference DAC 2026*; DAC '26; Association for Computing Machinery: New York, NY, USA, 2026.
- (2) Liu, S.; Yang, H.; You, X.; Luan, Z.; Liu, Y.; Qian, D. Efficient Locality-Aware Instruction Stream Scheduling for Stencil Computation on ARM Processors. In *Proceedings of the 39th ACM International Conference on Supercomputing*; ICS '25; Association for Computing Machinery: 2025; pp 250–264. <https://doi.org/10.1145/3721145.3725760>.
- (3) Liu, X.; Yang, X.; Ma, K.; Liu, S.; Zhang, K.; Yang, H.; Liu, Y.; Luan, Z.; Qian, D. Moirae: Generating High-Performance Composite Stencil Programs with Global Optimizations. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage, and Analysis*; SC '24; IEEE Press: Atlanta, GA, USA, 2024. <https://doi.org/10.1109/SC41406.2024.00026>.

Research Experience

High-Performance Sparse vDiT Engine for Video Diffusion Models (DAC 2026)Apr 2025 - Nov 2025 Beijing

BUAA & BYTEDANCE SEED AI RESEARCH

- First author
- Designed vDiT-SSR, a unified block sparse-attention format that composes static, semi-static, and dynamic 3D block patterns for video diffusion transformers.
- Built SPADE, a training-free sparse attention engine on vDiT-SSR that uses SICS-driven head-wise policies to generate per-input sparsity schemes and compile them into flash-style block-sparse CUDA/Triton kernels with low-overhead index search, operator fusion, and grouped launches.
- On Hunyuan-Video and Wan 2.1/2.2 text-to-video and image-to-video tasks, matched or improved generation quality while accelerating attention by 2.26x-3.40x and end-to-end throughput by 1.49x-1.80x over prior state-of-the-art sparse-attention methods.

C/C++ CUDA C/C++ Triton Python

ARM-Optimized Stencil Operator Compilation with Instruction Stream Scheduling (ICS 2025)May 2024 - Oct 2024 Beijing

SINO-GERMAN JOINT SOFTWARE INSTITUTE LAB, BUAA

- First author; source code: [AOSTencil Repo](#)
- Developed the SFTBR method (Serial-FMA to Tree-Based Reduction) for instruction reordering and instruction-level parallelism optimization.
- Designed a locality-aware scheduling model that leverages L3 cache, ccNUMA, and socket memory hierarchies.
- Integrated SIMD, tiling, and unrolling optimizations into a TVM-inspired domain-specific compiler.
- Achieved up to 4.39x speedup over existing stencil compilers on ARM platforms through GA-driven tuning and scheduling.

Python C/C++ ARM NEON

Globally Optimized Stencil Compilation for GPUs (SC 2024)Sep 2023 - Apr 2024 Beijing

SINO-GERMAN JOINT SOFTWARE INSTITUTE LAB, BUAA

- Built a data-movement-based cost model to predict GPU memory access patterns.
- Implemented horizontal (branching) and vertical (sequential) fusion optimizations for stencil operators.
- Developed an evolutionary search algorithm for computation-graph fusion guided by the cost model.
- Delivered 2.6x and 1.8x speedups over Open Earth and Artemis baselines, respectively.

Python CUDA C/C++

Internship Experience

Machine Learning Inference Optimization Engineer

BYTEDANCE SEED AI INFRASTRUCTURE TEAM

Feb 2025 - Present

Beijing

- High-performance Triton kernels and compiler support for NPUs
- ByteIR development and optimization for LLM workloads

C/C++ MLIR Triton Python

Project Experience

High-Performance Triton Kernel and Compiler for NPU

BYTEDANCE SEED AI INFRASTRUCTURE

Mar 2025 - Present

Beijing

- Hardened the Triton compiler by resolving defects uncovered while building an LLM operator library.
- Designed high-performance Triton operator implementations tailored for proprietary accelerator architectures.
- Built NPU kernels for LLM operators using processor-unit specialization, asynchronous data transfer, and compute-I/O overlap, delivering 2x speedup over the previous implementation.

C++ MLIR Triton Python

ByteIR Development and Optimization for LLM

BYTEDANCE SEED AI INFRASTRUCTURE

May 2025 - Aug 2025

Beijing

- Added LLM-focused features to **BYTEIR** and optimized GPU performance.
- Designed and developed ByteIR's new backend module **Triton-Backend** for efficient and flexible operator compilation.

C++ MLIR Triton Python

Dynamic Tensor Program Compilation Optimization for RISC-V Vector Extension

OPEN SOURCE SUMMER - BUDDY-COMPILER

Jun 2024 - Oct 2024

Beijing

- Implemented RISC-V optimization strategies with MLIR multi-level IR and Buddy Compiler backend support.
- Optimized operators including BatchMatmul, conv2d-nhwc-fhwc, and depthwise-nhwc-hwc.
- Achieved 12x speedup over native MLIR and 2x over the original Buddy-compiler through improved memory access patterns, vectorization, tiling strategies, and RISC-V vector instruction utilization.
- Leveraged RISC-V hardware features, including scalar-vector broadcast FMA and runtime-configurable vector register length.
- Integrated optimizations into TinyLlama and MobileNetV3, achieving up to 25% speedup on RISC-V platforms.
- Merged via **Buddy-mlir** and **Buddy-benchmark** pull requests.

MLIR C++

ARM Processor Optimization for Simulation Software

SINO-GERMAN JOINT SOFTWARE INSTITUTE LAB, BUAA

Mar 2023 - Sep 2023

Beijing

- Optimized sparse matrix and convolution operations for Phytium S2500 ARM CPUs.
- Designed a multi-level sparse memory access structure leveraging cache hierarchy.
- Delivered 20% speedup over ArmPL, Kokkos, and Halide baselines on Phytium S2500.

C/C++ OpenMP ARM NEON Python

Awards and Honors

Sep 2022	National College Innovation and Entrepreneurship Competition, Third Prize
Mar 2022	ASC 2022 World Student Supercomputer Challenge Finals (Team Leader), 6th / 300+
Apr 2021	National Programming Ladder Tournament Finals, Third Prize
Oct 2020	ICPC Northwest China Regional Contest, Silver Medal

Technical Skills

Technical Stack C/C++ | CUDA | Python | MLIR
English IELTS 6.5